

Designed by Remus Teodorescu 2026

LECTURE 12.2

Power Grid Stability Prediction

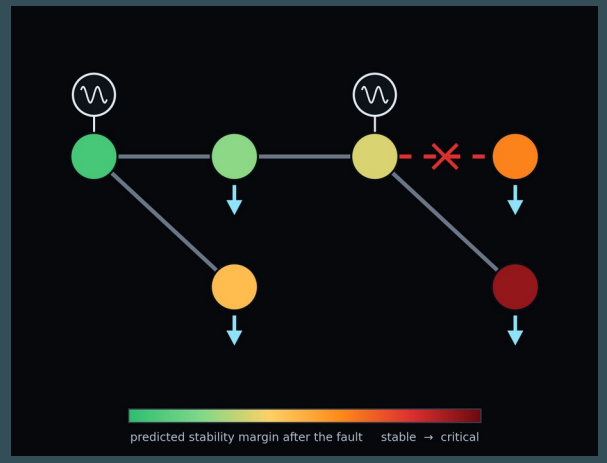
Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

FAULT · MARGIN · SCREEN



A line faults and the protection needs 140 ms to clear it: does the plant stay in step? The answer is the critical clearing time, one simulation per case, and the cases outgrow the time available. The last lecture screens them with a graph network and tests it on a topology it has never seen.

Learning Objectives

SECTIONS, IN ORDER

1. **Transient stability** — the clearing time, the scale, the margin
2. **The graph predictor** — an edge removed, and the honest test
3. **The screen in service** — microseconds, and what stays a simulation

UDL carries the graph network. The example is Ex_12.2's screening of the six-bus grid of the last lecture; the labels are the classical machine model's - no exciter, no governor, first swing only - and results inherit that.

READ, AND WHAT EACH ANSWERS

SciPy	how is a swing integrated?
UDL ch. 13	what does a graph layer share?
PyTorch	how is a model shipped?

The Question a Relay Asks

Critical clearing time — the longest a fault may stay before the machines fall out of step: the angle between them must stay under 180 degrees. The protection clears in 140 ms, so a case is insecure when its critical time is 140 ms or less - a verdict with a number behind it.

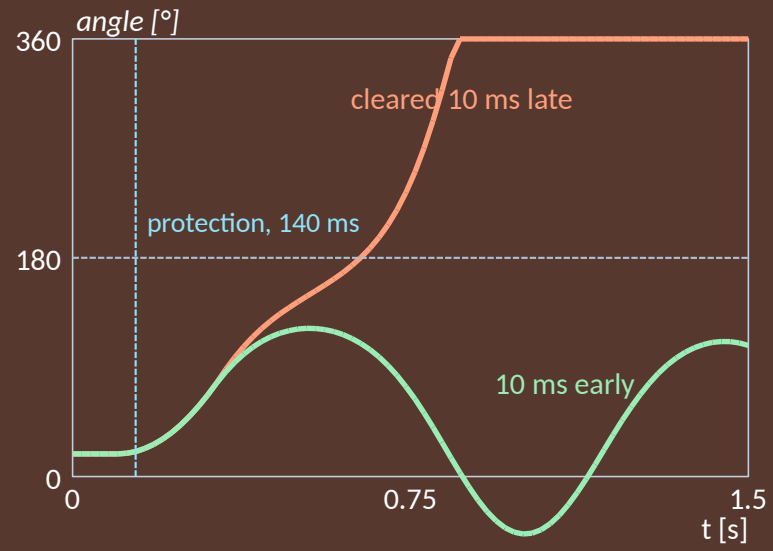
THE LABEL OF EVERY CASE

$$\mathrm{CCT} = \max\left\{t_c : \max_t |\delta_1(t) - \delta_0(t)| < 180^\circ\right\}, \qquad \text{insecure} \Leftrightarrow \mathrm{CCT} \le 140\ \text{ms}$$

t_c the clearing time, $\delta_1 - \delta_0$ the angle between the machines, 140 ms the protection's own time

Rotor-angle stability on the first swing: of the three stability questions - angle, frequency, voltage - this lecture asks the first, with the classical machine model supplying the labels.

THE TWO SIDES OF THE CRITICAL TIME



Line 0 out at the base point: the critical time is 220 ms, 80 ms beyond the protection's 140 ms, so this case is secure.

Screening Is Combinatorial

N-1 screening — the operator's standing question: for the current state and every single outage, would the grid survive? Every operating point times every contingency is a case, every case is a simulation in time with a bisection on top, and the cases outgrow the time available.

THE COUNT, ON SIX BUSES ALREADY

operating points 180, drawn around the base case

contingencies 6: each line out, one at a time

cases 1080, and 16 % of them insecure

The exercise's reference integrates all 1080 with Runge–Kutta at a 2 ms step and bisects each to 2 ms. A real grid has thousands of buses and a state that changes every few minutes, so the screen must be cheap and the simulation kept for the doubtful cases.

THE SIMULATION, AT THREE STEPS

step	worst CCT error	verdicts moved
2 ms	the reference	—
5 ms	4.88 ms	19 of 1080
10 ms	—	53 of 1080

The 5 ms step is the coarsest that meets the agreed 5 ms on the clearing time, at 0.6 ms a case batched; 16 % of the 1080 cases are insecure, and those are the ones a screen must not miss.

The Equal Area Criterion

Equal area criterion — stability read as energy on the power-angle curve: the machine stays in step if the area that can slow it after clearing is at least the area that sped it up during the fault. Their difference is a margin - how close a case is - where a verdict says only stable or not.

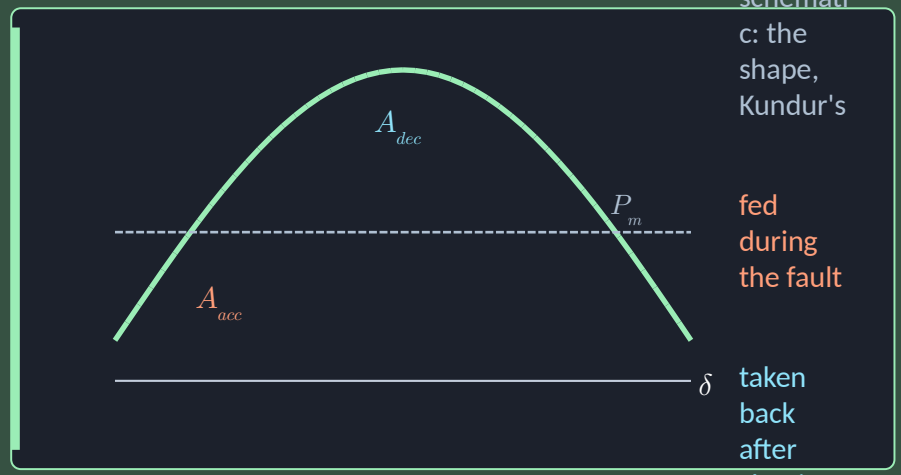
A VERDICT, AND THE NUMBER BEHIND IT

$$A_{acc} \le A_{dec}, \qquad \text{margin} = A_{dec} - A_{acc}$$

A_{acc} the energy the fault feeds the rotor, A_{dec} what the grid can take back after clearing : read on the $P(\delta)$ curve

TALK · 2 MIN How would you explain, in your own words, why an operator wants a margin rather than a verdict? Then, in the drawing on the right, what happens to the two areas when the fault is cleared later?

THE TWO AREAS, ON THE CURVE



schematic: the shape, Kundur's

fed during the fault

taken back after clearing

in Ex_12.2 the margin is CCT − 140 ms

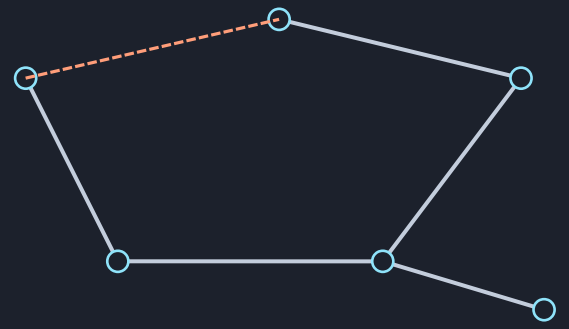
so the screen ranks cases, not just flags them

An Edge Removed

Contingency as a graph change — a tripped line is an edge removed, not a new value in a feature vector. A dense network's inputs are a fixed list, so a missing line is outside what it can represent; a graph network applies one local rule on whatever graph it is given.

THE SAME BUSES, A DIFFERENT GRAPH

LINE 0 OUT, AS THE NETWORK SEES IT



the same buses, one edge gone

not a new value: a new graph

the local rule still applies

WHAT GOES IN, WHAT COMES OUT

node features	the operating point at each bus
edge features	each line's admittance
the graph	with the outage applied
out	the clearing time of the case
weights	6721, shared over the graph

The same drawing convention as the last lecture's grid: this is Ex_12's own wiring, one edge gone.

Message Passing

Message passing — the graph network's one move: every bus gathers a message from each neighbour, through the line that joins them, and updates its own features - the same two small networks at every bus and line, on whatever graph is given.

ONE LAYER, WRITTEN OUT

$$h_i' = \phi\left(h_i, \sum_{k \in N(i)} \psi(h_i, h_k, e_{ik})\right)$$

h_i bus i's features, $N(i)$ its neighbours on THIS graph, e_{ik} the line's features, ϕ, ψ the same small networks everywhere

ONE GRAPH LAYER IN PYTORCH

```
psi = MLP(n_in=2*F+E, n_hidden=32, n_layers=2)   # the message, per line
phi = MLP(n_in=2*F,   n_hidden=32, n_layers=2)   # the update, per bus
msg = psi(h[src], h[dst], e).scatter_sum(dst)    # gather over THIS graph's edges
h   = phi(h, msg)                                # the same weights at every bus
```

MEASURED, 216 HELD-OUT CASES

	5 ms sim	GNN
mean CCT error	1.66 ms	0.57 ms
insecure missed	0 of 26	0 of 26
false alarms	5	0
one case	0.6 ms	3 µs
training	—	11 s

Held-out operating points of topologies seen in training: the friendly test. No dense network is trained here, so the notebook makes no claim against one - that comparison is an Advanced Topics row.

Held-Out Topologies

The honest test — held-out samples of topologies seen in training measure interpolation, and the rare insecure cases are exactly the ones that must be right. The fair test holds out a topology: train with line 0's outage never shown, then ask about it. The exercise does, and reports what happens.

WHAT THE EXERCISE MEASURES

seen in training	line 0 out: 0.69 ms error, none missed
never seen	57.21 ms error, all on the unsafe side
the verdicts	11 of 11 insecure cases missed



grey: predicted = true

orange: shifted, unsafe side - the order survives

TALK · 2 MIN How would you decide, in your own words, what a fair test of this screen is? Then, in the drawing above, why does a constant shift keep the ranking - and why was finding the 185 ms a cheat?

READ IT PLAINLY

the error	a nearly constant offset
so	the ranking of cases survives
but 185 ms	was found with the truth in hand
so in service	unseen topologies get simulated
the screen	orders the queue, nothing more

The predictor screens; it is not a tool of record, and the slide says exactly what was measured.

Microseconds and Their Place

Screening in service — where the trained screen would live and what stays a simulation. At three microseconds a case the network orders the whole queue in milliseconds; the simulation, the tool of record, is spent on the doubtful and the unseen. Ex_12.2 times both.

THE DIVISION OF LABOUR

the screen	every case, every few minutes, locally
the simulation	the doubtful cases, and every unseen topology
the record	always the simulation's, never the screen's
retraining	when the grid itself changes

THE TIMES, SIDE BY SIDE

	one case
reference, 2 ms step	1.7 ms
simulation, 5 ms step	0.6 ms
graph network	3 µs
all 1080, batched	1.7 s reference

Batched simulation is itself fast here: the network's case is the real grid, where a case is seconds, not milliseconds - and the screen's milliseconds leave the budget to the physics.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
the dense baseline	what structure wins against, measured	UDL ch. 13	the claim the deck does not make
exciters and governors	control acting within the first swing	ANDES docs	beyond the classical labels
ANDES, a tool of record	the labels checked against a full simulator	ANDES docs	Ex_12.2 section 5
frequency and voltage stability	the two questions not asked here	ENTSO-E	low-inertia and weak grids
one-shot against rollout	the verdict at once, or the states marched	Liu ch. 5	longer horizons

WHY THEY ARE LEFT OUT

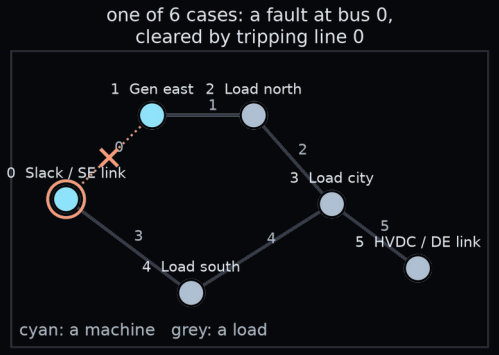
Ex_12.2 screens one grid's 1080 cases against its own Runge–Kutta reference, so none of them is needed to do it. The dense baseline stays an open measurement.

EXERCISE

Exercise 12.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the contingencies, the swing, one case 10 ms either side
- **4** — the reference on all 1080 cases; the step study
- **5** — the worst case in ANDES, classical and detailed
- **6** — the graph layer and the loss; 144 points train, 36 test
- **7** — line 0 held out entirely; the threshold raised until safe



THE QUESTION EACH PART ANSWERS

1 – 4	what is insecure, at what step?
5	does a full simulator agree?
6	can a screen match the labels?
7	what happens off the map?

REPORT QUESTION, TO CONCLUDE

On the unseen outage the screen was 57 ms wrong, on the unsafe side, and still ranked the cases. Write the operating rule you would give a control room for this screen: when is it trusted, and when is it ignored?

SUMMARY

Key Takeaways

TRANSIENT STABILITY	
what it is	do the machines stay in step?
function	a clearing time per case
train, tune	simulated labels, bisected
vs a verdict	a margin: CCT − 140 ms
used for	the queue of N−1 cases

GRAPH PREDICTOR	
what it is	message passing on the outage graph
function	the clearing time, in 3 µs
train, tune	on simulations; 6721 weights
vs held-out samples	topologies are the fair test
used for	ordering the work, never the record

Next — L13: the mini projects, where one of these problems becomes yours.

Questions

1 – 2 · LEARNING THE CLEARING TIME

1 **Why is a network trained to predict the critical clearing time of each case rather than the verdict stable or not?**

Every operating point times every outage is a case, and each label costs a simulation of the swing equation in time, eight with the bisection, so the cases outgrow the time available. A network trained once on simulated cases answers a new one in a few microseconds. The clearing time minus the protection time of 140 ms is a margin: it says how close a case is to the limit, where a verdict says only stable or not, so the cases can be ranked. *Kundur ch. 13; Ex_12.2 sections 3 – 4*

2 **Where is the physics in the graph network of the exercise, if not in its loss?**

The loss is only the mean squared distance between the predicted and the simulated critical clearing time. The physics is in the labels, which come from the swing equation integrated in time with the classical machine model, and in the graph the network is handed, the grid with the outage applied. The network therefore learns how the clearing time varies with load and import on the graphs it has seen; it does not solve the swing equation. *UDL ch. 13; Ex_12.2 section 5*

3 – 4 · THE GRAPH AND THE UNSEEN LINE

3 **How does a graph network take an outage as input, and what does that allow that a dense network does not?**

A tripped line is an edge removed from the graph of the grid, not a new value in a list of inputs. Each graph layer combines the numbers at every bus with those of its neighbours along the lines that remain, with the same weights at every bus, so the network can be applied to any graph it is given. A dense network takes a fixed list and cannot represent a missing line. No dense network is trained in the exercise, so no comparison of accuracy is claimed. *UDL ch. 13; Ex_12.2 section 5*

4 **Why is a held-out topology the fair test of the graph network, and what did that test show in the exercise?**

Held-out operating points of outages seen in training measure interpolation: there the network was within about 0.6 ms and missed no insecure case. Trained without line 0 out and then asked about it, it was 57 ms off on the unsafe side and missed all eleven insecure cases. Its error was nearly constant, so the ranking held, but the threshold that rescued it was found with the truth in hand. So the network orders the queue, and every unseen topology goes to the simulation. *Ex_12.2 section 6*

NEXT

L13 — The Mini Projects

One month, one problem of your own, chosen from the Part 2 sets: thermal, flow, battery, solid oxide, the car, or the grid - and the Thor runs the screen.

BEFORE THE NEXT SESSION

read	UDL ch. 13
run	Ex_12.2, sections 1 to 6
bring	your operating rule for the screen
choose	the mini project you will take

Part 2 is finished: heat, flow, electrochemistry, a car, and a grid - every one a physics, a classical answer, and a network earning a stated place beside it.